

KAREENA LAKHANI

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Professional Summary

AI/ML engineer focused on agentic systems, multimodal reasoning, robotics perception, and autonomy. Experienced in building LLM/VLM agent frameworks, retrieval-augmented memory systems, deep learning experiments, and robotics pipelines using Python, PyTorch, ROS2, LangChain/LangGraph, and AWS. Strong research-to-implementation background with measurable project outcomes across visual reasoning, locomotion, and visual odometry.

Technical Skills

AI/ML :	PyTorch, LangChain, LangGraph, Retrieval Augmented Generation(RAG), VECtor Databases(FAISS, Chroma), Large Language Models(LLMs), Vision Language Models(VLMs), Natural Language Processing, Computer Vision, Machine Learning, Deep Learning, Neural Networks, Reinforcement Learning, Agentic AI, OpenCV
Robotics & Autonomy:	ROS/ROS2, Gazebo, SLAM, Motion Planning, Path Planning, Sensor Fusion
Programming & Tools:	Python, C/C++, MATLAB, Linux, Docker, Git, Shell Scripting
Cloud & APIs:	AWS Lambda, API Gateway, S3, DynamoDB, AWS Bedrock, Anthropic Claude API
Hardware :	MyCobot Pro 600, Raspberry Pi, Atmega, Arduino

Education

Arizona State University, Tempe, AZ	Aug 2024 - May 2026
M.S. Robotics & Autonomous Systems (AI)	GPA: 3.90/4.00
MIT-ADT University, Pune, India	Aug 2019 - Jul 2023
B.Tech. Electronics & Communication Engineering	GPA: 8.60/10.00 (3.54/4.0 US)

Selected Projects

Multimodal VLM Agent With Semantic Memory and Context Engineering (Aug 2025 - Dec 2025)

- Built a multimodal VLM agent framework with ReAct-style reasoning, OCR tool use, semantic memory retrieval, and episodic state tracking using Python, PyTorch, Sentence-Transformers, and transformer-based architectures.
- Implemented retrieval-augmented memory with MiniLM embeddings and cosine-similarity loop detection to support long-horizon reasoning across noisy visual streams.
- Improved Kirby benchmark progress from 5% to over 20% by redesigning short-term and episodic memory buffers for better state continuity.
- Evaluated the framework on VideoGameBench-Lite across Kirby, Doom II, Pokemon Crystal, and Super Mario Land, measuring task completion, loop reduction, and token efficiency.

NeuroEvolution of Augmenting Topologies (NEAT) for Bipedal Locomotion Jan 2026 - May 2026

- Evolved neural network controllers for BipedalWalker-v3 using NEAT, achieving a best average reward of 272.92 over 20 evaluation episodes with a 0% failure rate.
- Designed and compared three fitness functions (speed-only, energy-efficiency, stability-aware); stability shaping with hull tilt and angular velocity penalties outperformed baselines, showing reward design drives gait quality over complexity alone.
- Ran a fixed-topology ablation confirming that structural mutation contributed 48.73 reward points, validating that topology evolution – not just weight optimization – is critical for escaping locomotion local optima.
- Implemented and evaluated curriculum learning across progressive terrain stages; showed early acceleration without hardcore transfer, attributing the gap to catastrophic forgetting of flat-terrain gait representations.

Deep Patch Visual Odometry (DPVO) – Ablation Study Jan 2025 - May 2025

- Conducted an ablation study on DPVO architecture evaluating 4 modifications – Euclidean distance, ConvLSTM, Softmax temperature tuning, and Gradient-based Temporal Attention (GTA) – on the TartanAir dataset.
- Benchmarked trajectory accuracy using Absolute Trajectory Error (ATE) across 16 sequences on an NVIDIA RTX 4090, replicating published DPVO results within a few centimeters.
- Identified Softmax temperature tuning as the best-performing variant, achieving the lowest average ATE of 0.21m across all tested sequences.

Experience

Software Analyst Intern, Capgemini, India Feb 2023 - Jun 2023

- Developed comprehensive test plans, test cases, and review templates for web application workflows, improving test coverage, documentation quality, and requirement traceability across functional modules.
- Automated end-to-end test scenarios using Selenium WebDriver, Java, TestNG, Maven, XPath/CSS selectors, and Page Object Model, reducing manual regression effort by an estimated 30-40% and improving script maintainability.
- Performed API validation using Postman and executed functional/regression test cases within a hybrid automation framework, identifying defects early and supporting faster QA feedback cycles for the development team.